Ahmed **Cheikh Rouhou**

Engineering Manager · PhD in Computer Science

Ariana - Tunisia

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Summary

Engineering Manager leading an AI team of up to 20 through senior technical leads, with a PhD in Computer Science and 10+ years across applied research and production machine learning. Delivered computer vision, NLP and computational pathology systems from research to production, with QA validation for GxP-regulated environments. Hands-on background in Transformers, graph neural networks and MLOps, and author of 10+ peer-reviewed publications. Builds teams and talent through hiring, internship programs and university collaborations, and uses LLM agents and MCP tooling to streamline engineering workflows.

Skills

Leadership	Team and team-lead management, technical hiring, internship programs, sprint planning, SOWs, GxP validation
AI & ML	PyTorch, TensorFlow, Transformers, Computer Vision, NLP, Graph Neural Networks, Foundation Models, AlphaFold, Hidden Markov Models (HTK, Kaldi)
LLM Agents	Claude, AI agent workflows, MCP servers and agent skills for integrating agents into team workflows
MLOps & Cloud	Docker, Kubernetes, cloud platforms, CI/CD, MLflow, Weights & Biases, ONNX
Coding	Python, C/C++, Java/JEE, SQL, Bash
Databases	PostgreSQL, MySQL, Oracle, HDF
Languages	Arabic, French and English

Experience

InstaDeep Ltd — *Tunis, Tunisia*

Engineering Manager — *Jan. 2026 - Present*

- Lead a team of up to 20 people, including interns, organised around senior technical leads; define hiring plans and co-create statements of work (SOWs).
- Own quality assurance and validation of AI projects for GxP-regulated environments.

Senior Research Engineer — *Sep. 2023 - Dec. 2025*

- Project lead for histology (H&E) whole-slide image segmentation: owned architecture and sprint planning while hands-on. Integrated **LongViT** as backbone (images up to 32,768 × 32,768 px), improving on the baseline, and released production models deployed through ONNX.
- Managed three team leads and a team of 11. Ran the internship program end to end (12+ interns recruited) and led collaborations with 5+ Tunisian universities, teaching Transformers, LLMs and computer vision, and creating hackathons.

Research Engineer & Team Lead — *Sep. 2021 - Sep. 2023*

- Led a team of 4 (engineers and interns). Researched AI-enabled peptide design with a biotech partner using Transformers, graph neural networks and AlphaFold, focused on peptide binding prediction.
- Co-supervised a CRNS PhD thesis on continual learning for handwritten-document foundation models (MSdocTr-Lite, UCL-MHTR). Led an internship project combining Transformers and GCNs, state of the art on IAM and ICDAR 2017 IE (WACV 2024).

Research Engineer — *Jan. 2020 - Sep. 2021*

- Designed and implemented (PyTorch) a Transformer for joint handwriting and named entity recognition, the first of its kind and state of the art on ICDAR 2017 IE (*Pattern Recognition Letters*, 2022), with CRNS.
- Contributed to pre-training and data preparation of **TunBERT**, the first open-source BERT for the Tunisian dialect (NVIDIA NeMo, multi-GPU), with NVIDIA and iCompass; code and datasets released, two publications, NVIDIA GTC talk.

Digital Research Centre of Sfax (CRNS) — *Sfax, Tunisia*

Researcher & PhD Student — *Mar. 2015 - Dec. 2019*

- Word spotting and indexing of printed and handwritten multi-script documents (thesis topic); five publications.

Esprit School of Business, University of Sfax, Training Pro Group — *Tunisia*

Lecturer & Instructor — *2017 - 2023*

- Taught machine learning and deep learning fundamentals, neural networks for image classification, JEE, Spring, Hibernate, Perl and Python.

Logicom Company — *Sfax, Tunisia*

Full Stack Developer — *Sep. 2013 - Jan. 2016*

- Built real-time synchronization (RMI, FTP, SQL) and backup modules; upgraded the product to .NET (C#, WPF, NHibernate) and added device integrations (RFID, SMS, time attendance).

Talan Tunisie Internationale — *Tunis, Tunisia*

Full Stack Engineer — *Feb. 2013 - Jul. 2013*

- Created an API for a geographic information system and a web application for marking construction sites.

Education

ENIS (National School of Engineers of Sfax) — *Sfax, Tunisia*

PhD in Computer Science — *2015 - Jun. 2021*

- Thesis: *Keyword Spotting in Printed and Handwritten Multi-Script Document Images*; defended with honors. Supervisors: Yousri Kessentini, Slim Kanoun.

FST (Faculty of Sciences of Tunis) — *Tunis, Tunisia*

Computer Science Engineer — *Sep. 2010 - Aug. 2013*

Selected Publications

2025	**UCL-MHTR: A unified continual learning system for multilingual handwritten text recognition**, Expert Systems with Applications	*Link*
2024	**Graph Neural Networks for End-to-End Information Extraction From Handwritten Documents**, WACV	*Link*
2023	**MSdocTr-Lite: A lite transformer for full page multi-script handwriting recognition**, Pattern Recognition Letters	*Link*
2023	**TunBERT: Pretraining BERT for Tunisian Dialect Understanding**, SN Computer Science	*Link*
2022	**Transformer-Based Approach for Joint Handwriting and Named Entity Recognition in Historical documents (first author)**, Pattern Recognition Letters	*Link*
2021	**Multi-task learning for simultaneous script identification and keyword spotting in document images (first author)**, Pattern Recognition	*Link*
2020	**Hybrid HMM/BLSTM system for multi-script keyword spotting in printed and handwritten documents (first author)**, Neural Computing and Applications	*Link*

12 publications in total; full list on Google Scholar.

Certifications

NVIDIA

NVIDIA Deep Learning Institute: University Ambassador — *2019 - 2025*

- Efficient LLM Customization and Deploying RAG Pipelines at Scale (2025); Generative AI with Diffusion Models (2023); Transformer-Based NLP Applications and Fundamentals of Deep Learning, including Multi-GPU and Computer Vision (2019 - 2022).

Udacity

Machine Learning DevOps Engineer Nanodegree — *2022*

Community & Committees

Ongoing	**Judge, Mentor & Technical Advisor**, AI hackathons, workshops and community events, including Deep Learning Indaba and IndabaX (3+ per year)	*Tunisia*
2023	**Publicity Chair**, International Conference on Intelligent Systems and Pattern Recognition (ISPR)	*MENA*
2023	**Guest Lecturer**, Transformer-Based NLP, TICV Master's program, ENIT	*Tunisia*
2020	**AI Trainer & Speaker**, DSC Academy and DSC SupCom (Data Science)	*Tunisia*
2019	**Organizer & Teaching Assistant**, Deep Learning IndabaX Sfax and NVIDIA Deep Learning Spring School; BioDialog Summer School (2018)	*Tunisia*